How to measure wheat seed shape

If the number of images is large, you may exhaust Google Colab's free quota. In that case, either purchase Compute units or download the script to your local environment and run it in Jupyter Lab. With 100 Compute units, you can analyze thousands to tens of thousands of images; upload speed will be the rate-limiting factor.

① Sample data

② Print the template sheet and measure scale

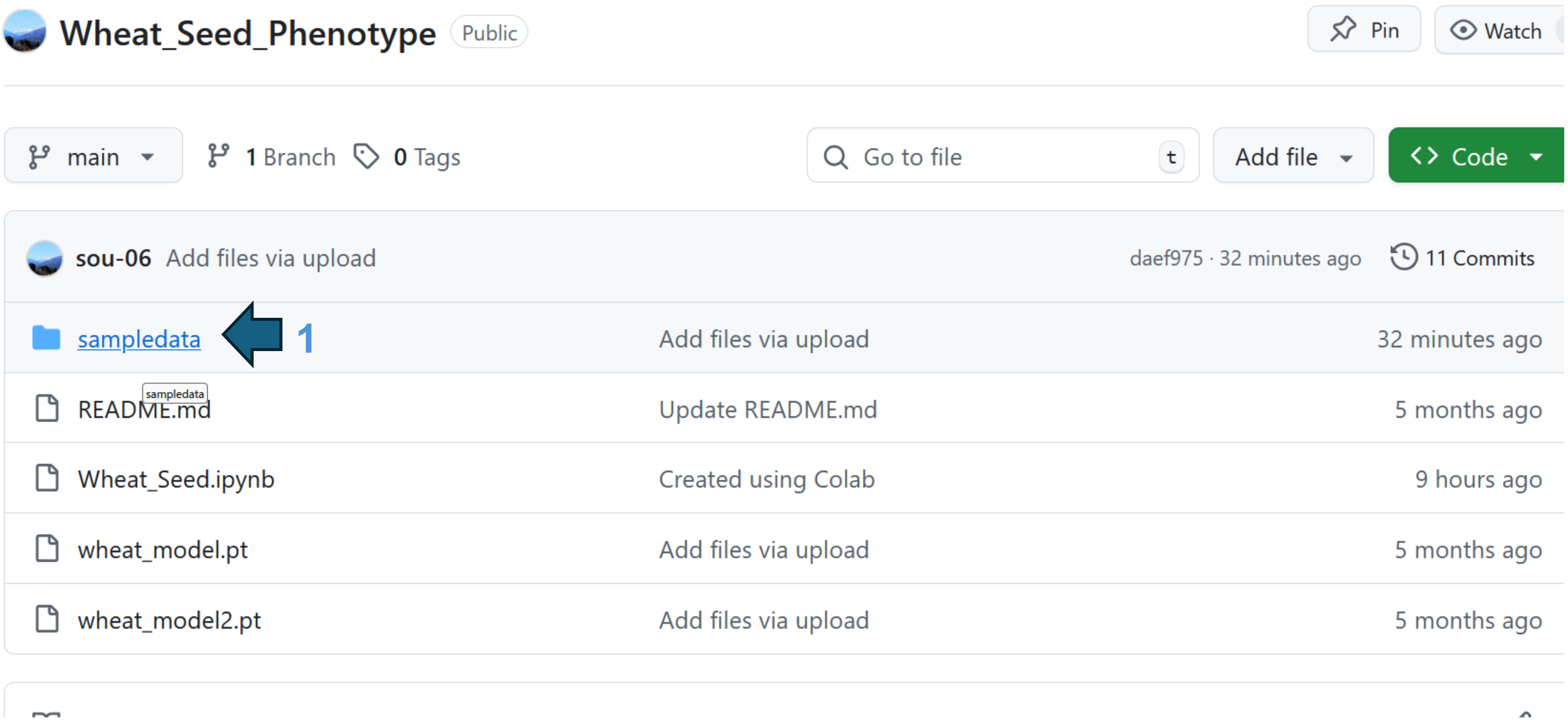
③ Take picture

④ Run the script on google drive

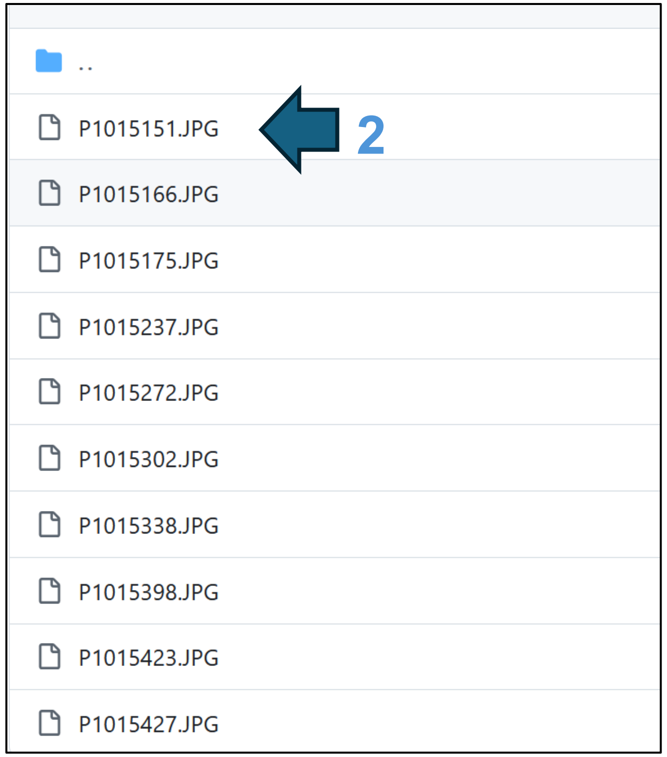
④ data check and analyze

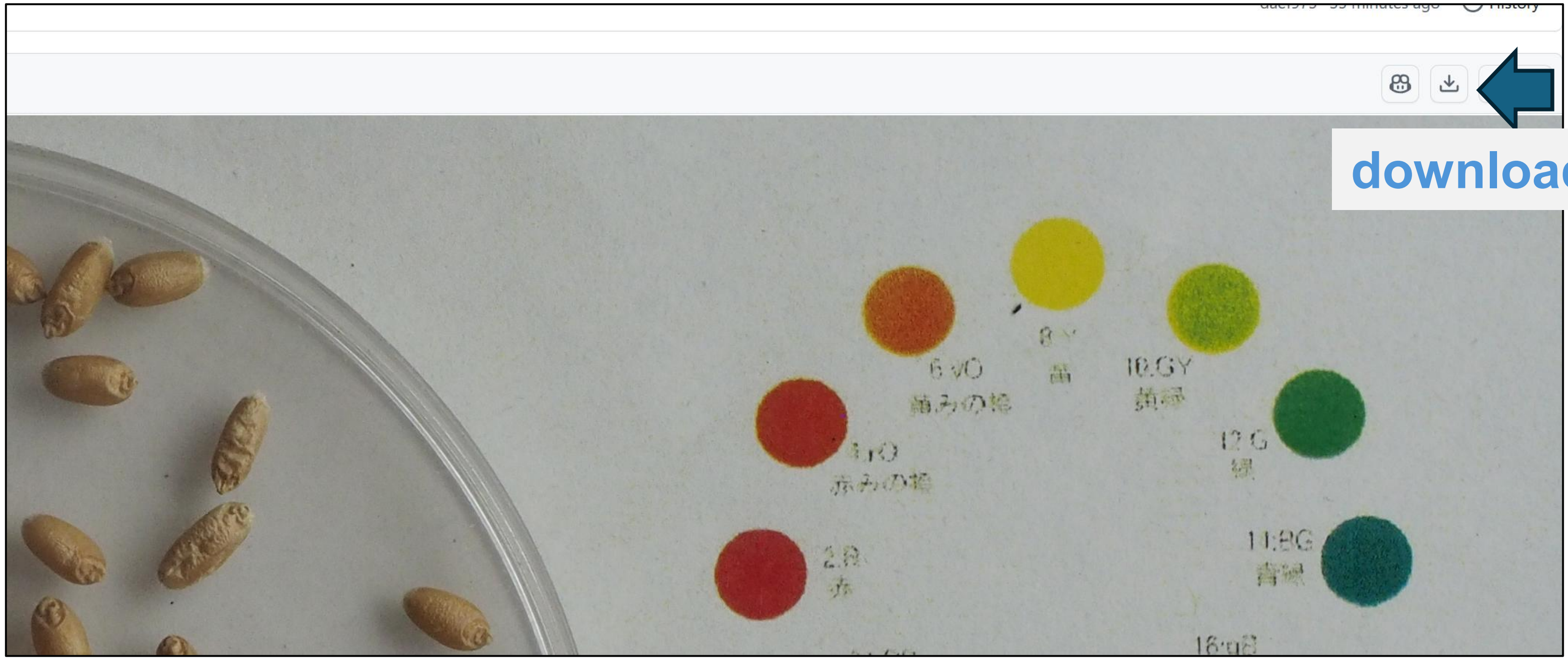
① Sample data

Access to
https://github.com/sou-06/Wheat_Seed_Phenotype/tree/main?tab=readme-ov-file



① - 1. Download sample data.



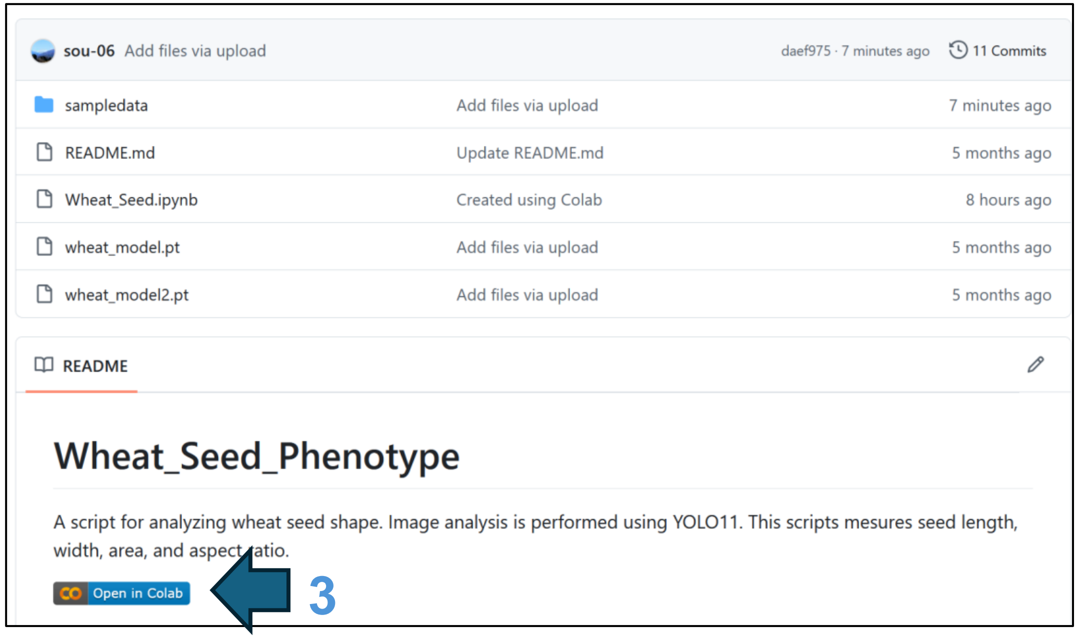


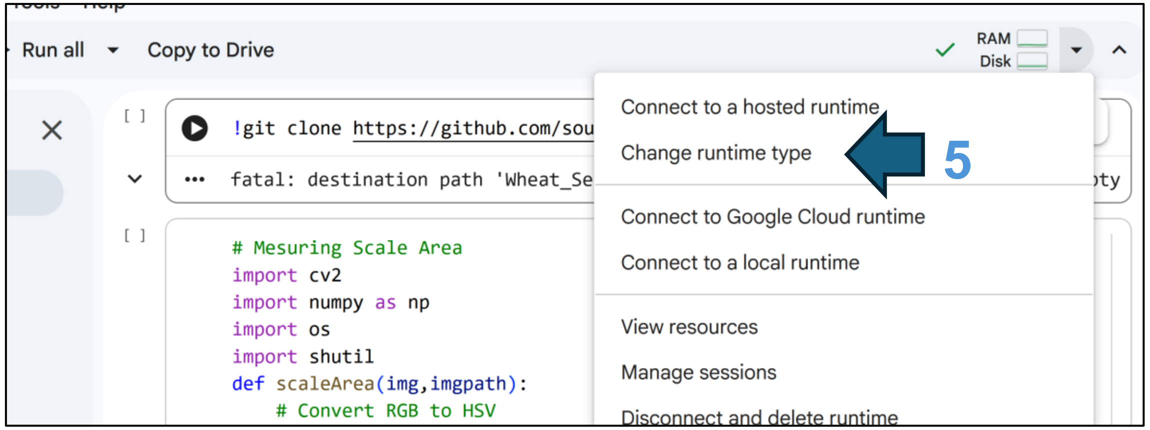
Repeating similar operations to download several different images (3 – 10 pictures).

① - 2. Upload sample data

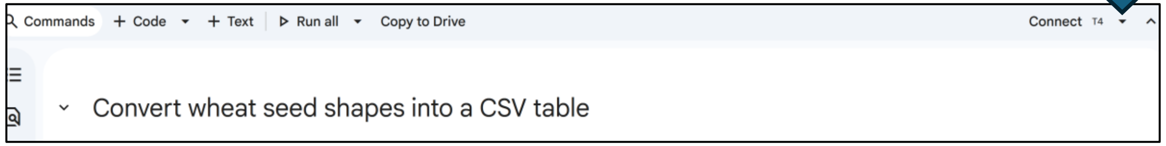
Access to
https://github.com/sou-06/Wheat_Seed_Phenotype/tree/main?tab=readme-ov-file

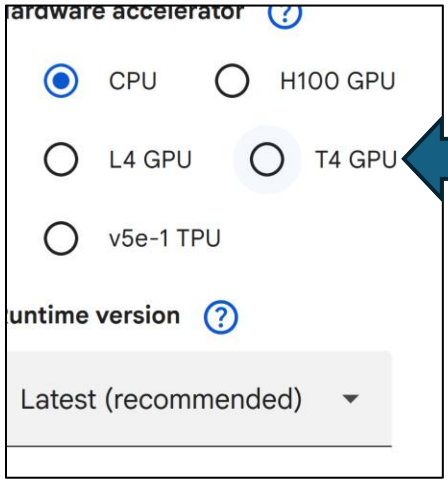
(2) Open google colab





4

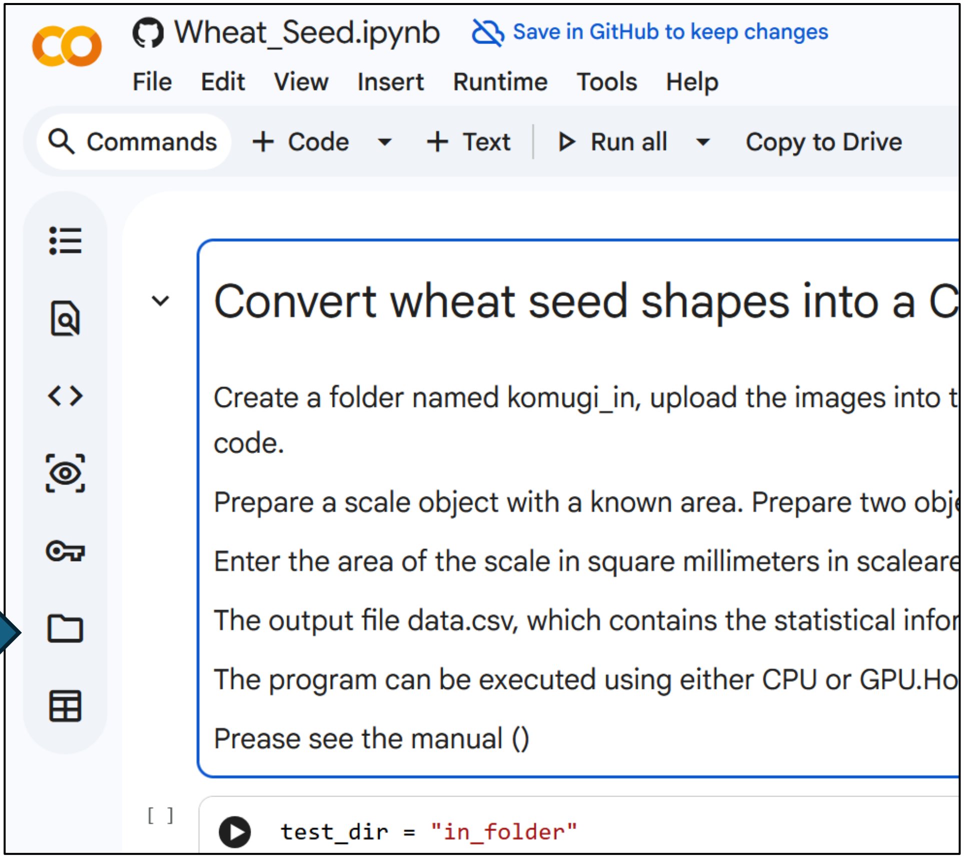




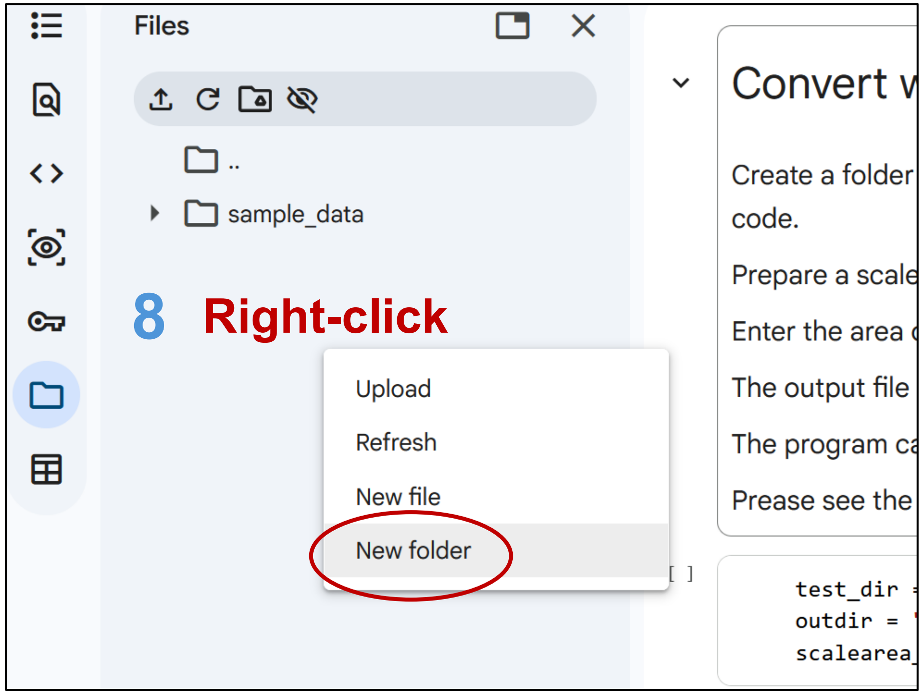
6. Choose T4 GPU ->Save

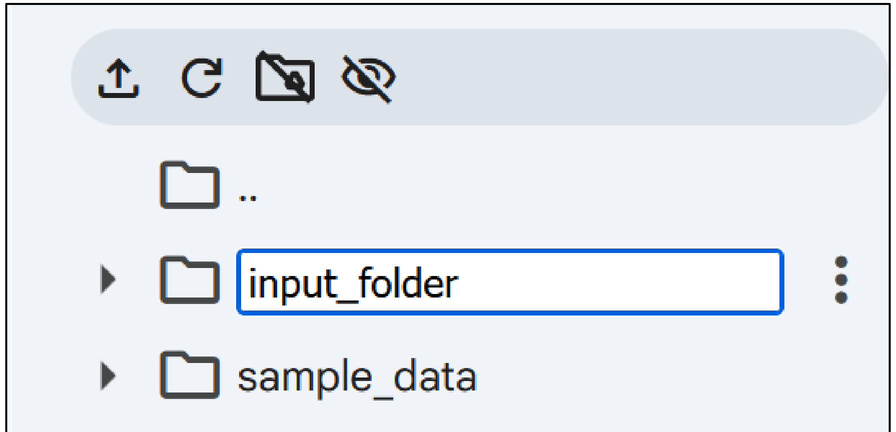
① - 2. Upload sample data

7



8 **Right-click**

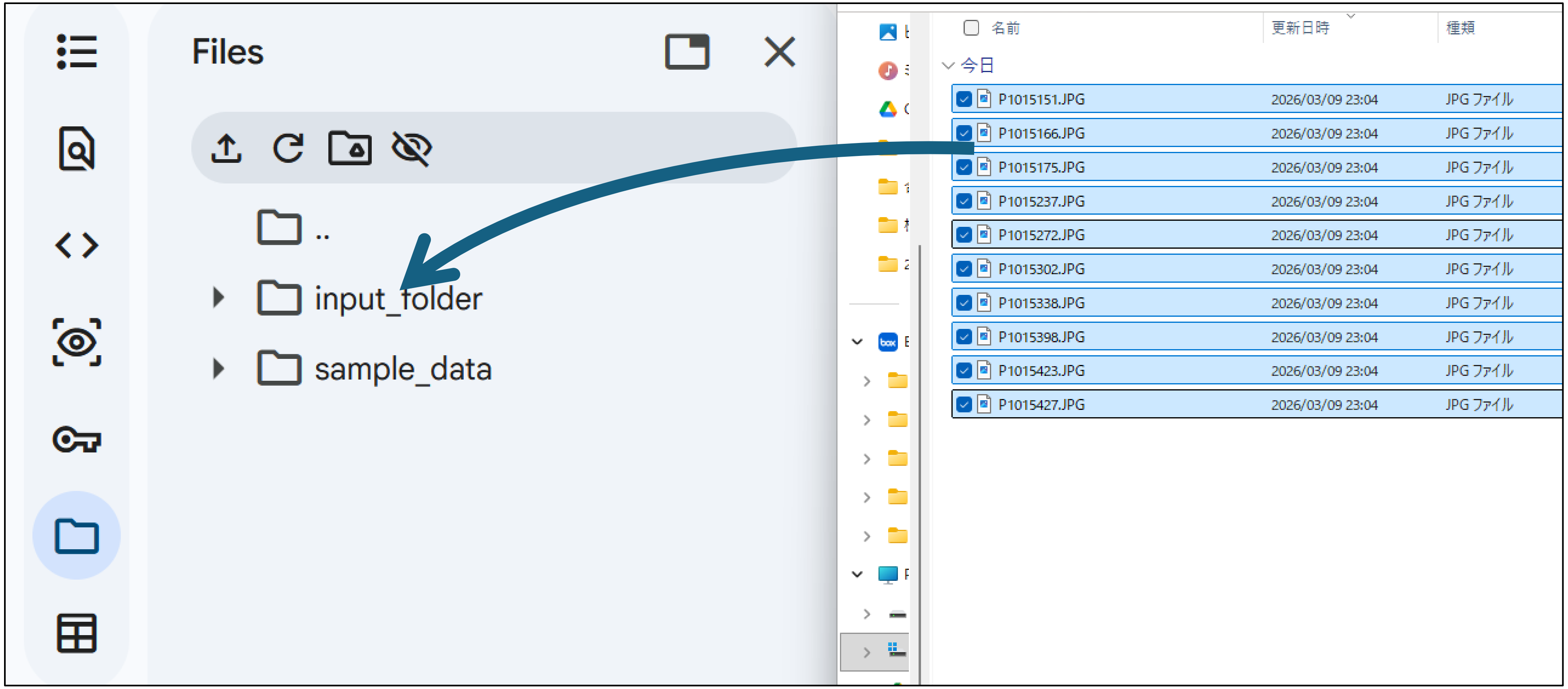




9 **(Spelling is important)**
Change folder name to “input_folder”

① - 2. Upload sample data

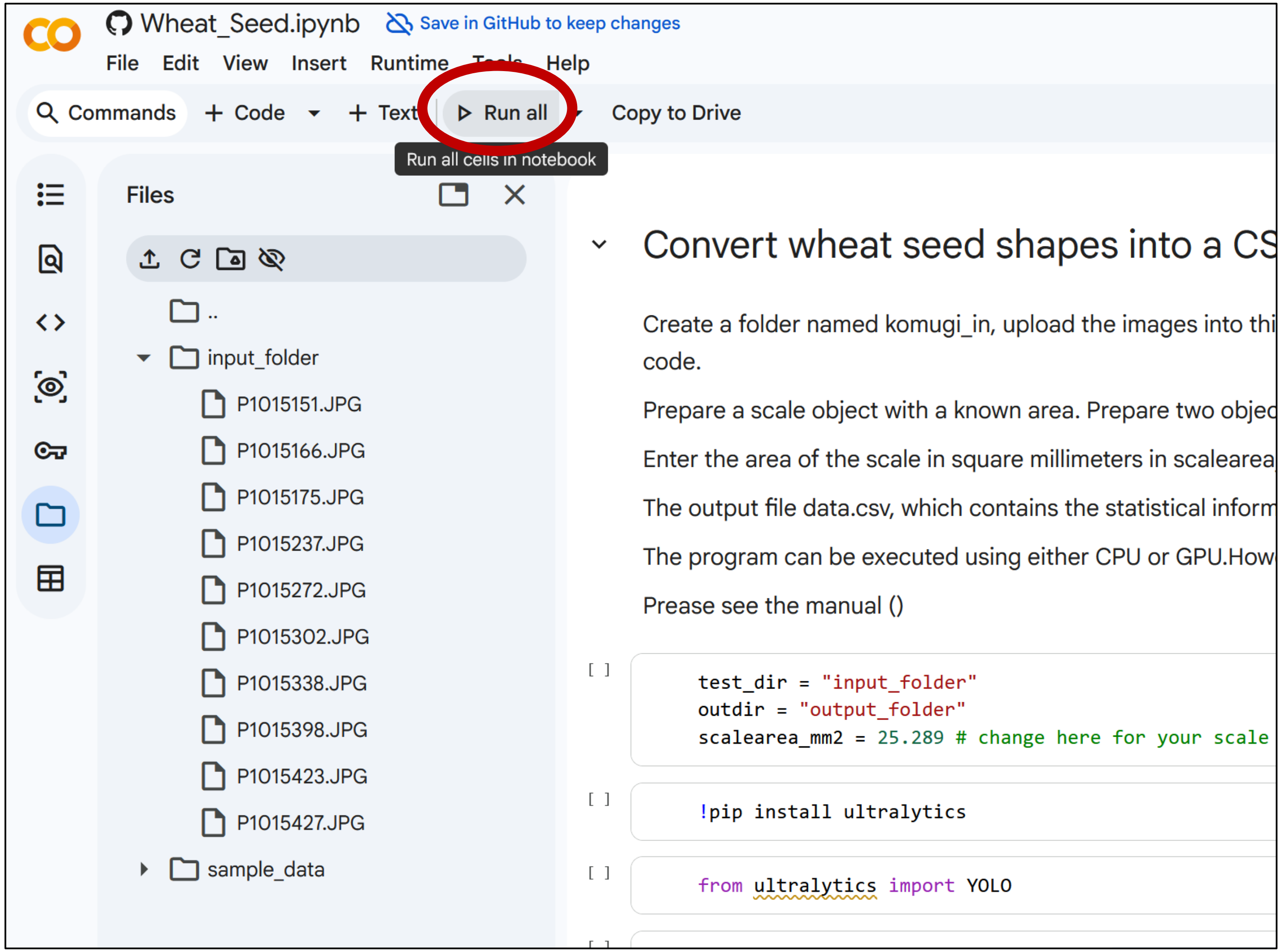
10. Drag and drop pictures into “input_folder”



① - 3. Run sample data

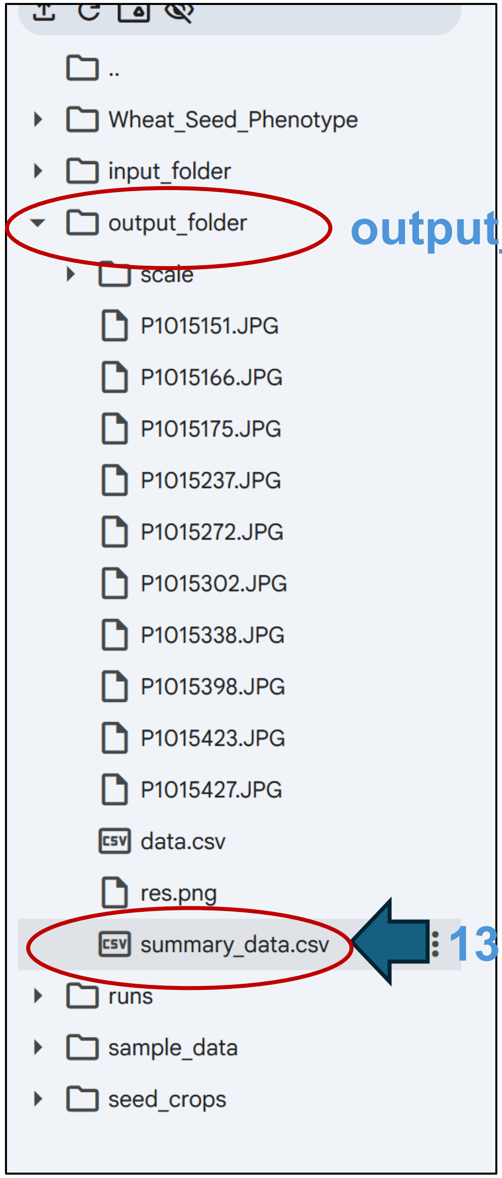
11. ▷ Run all

Please wait few minutes.



① - 4. Check result

12. Scroll down and wait for the cell for “Making summarized data” to complete.



output_folder

13. Right-click and download

Output_folder > **sunnary_data.csv**

① - 4. Check result

	length_mm	length_mm	length_mm	length_mm	width_mm	width_mm	width_mm	width_mm	Area_mm2	Area_mm2	Area_mm2	Area_mm2
	mean	std	median	count	mean	std	median	count	mean	std	median	count
File_Name												
P1015151	6.291882	0.537137	6.344463	90	3.071284	0.253556	3.051417	90	15.21821	2.126726	15.30431	90
P1015166	6.545488	0.480854	6.595992	165	3.230528	0.289725	3.260713	165	16.66667	2.356914	16.8883	165
P1015175	5.974785	0.440631	5.980284	89	3.044792	0.302523	3.054924	89	14.36147	2.226711	14.52029	89
P1015237	5.780698	0.371865	5.804195	93	3.142547	0.290471	3.139018	93	14.32659	2.066813	14.47962	93
P1015272	5.888534	0.475832	5.875616	57	3.113154	0.34306	3.142855	57	14.48763	2.491514	14.50333	57
P1015302	5.961168	0.414797	5.936583	100	3.332691	0.320769	3.317596	100	15.65203	2.198473	15.55513	100
P1015338	5.89249	0.439844	5.89982	126	2.799051	0.386438	2.793023	126	13.02064	2.394226	13.08756	126
P1015398	6.269714	0.487978	6.331117	63	2.804117	0.408462	2.789094	63	13.87419	2.608743	14.25247	63
P1015423	6.088394	0.468615	6.14058	64	2.817258	0.33201	2.718471	64	13.53726	2.303346	13.40427	64
P1015427	6.247933	0.392415	6.316151	87	3.128766	0.353329	3.143161	87	15.42752	2.436478	15.86754	87

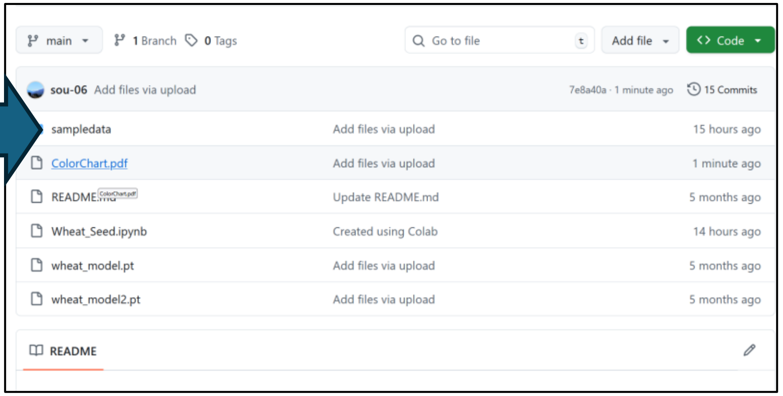
**For each image, lengths and widths are calculated in millimeters.
Count is the number of seeds detected**

② Print the template sheet and measure scale

② Print the template sheet and measure scale

The size of the markers varies depending on the printer, so the marker size should be set each time printing. **For comparative data, use the same paper or consecutively printed sheets and apply the same scale values. When using the same paper, please use the same value without remeasuring.**

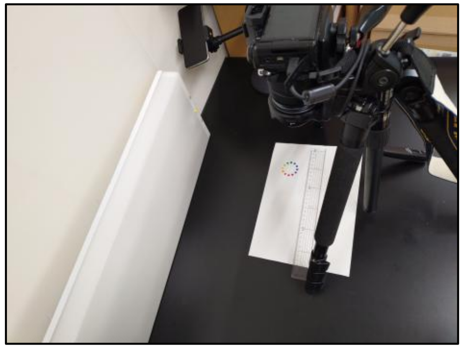
1

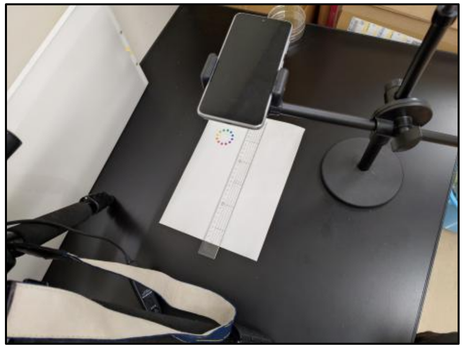


2. Download and printout

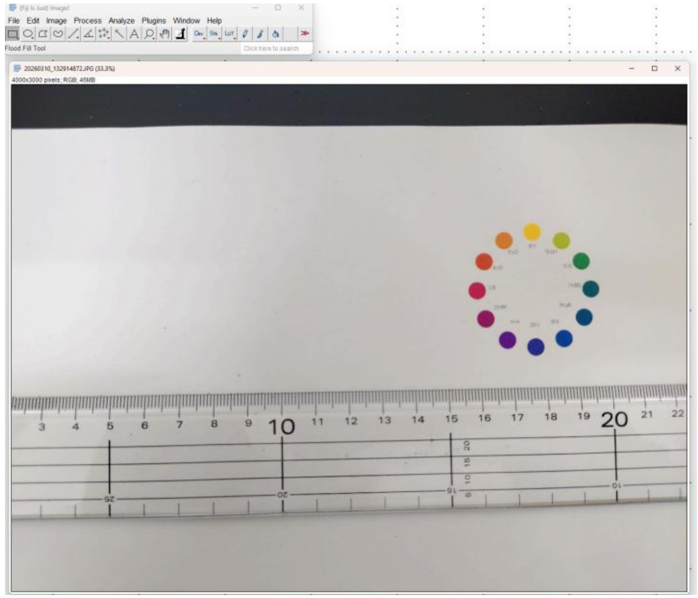
3. Take picture with measure.

Make sure the camera is level.

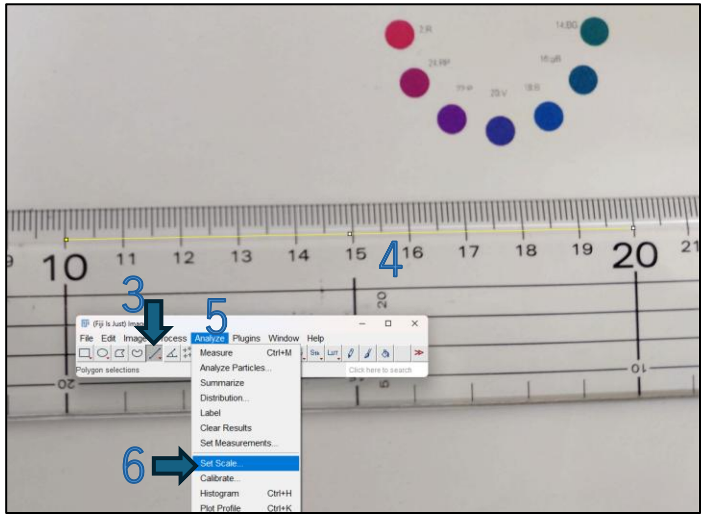




② Print the template sheet and measure scale



2. Open with ImageJ. (Fiji)



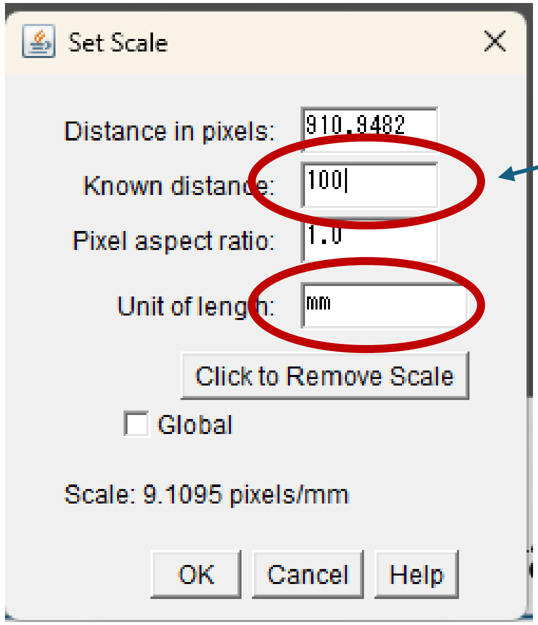
3. Straight

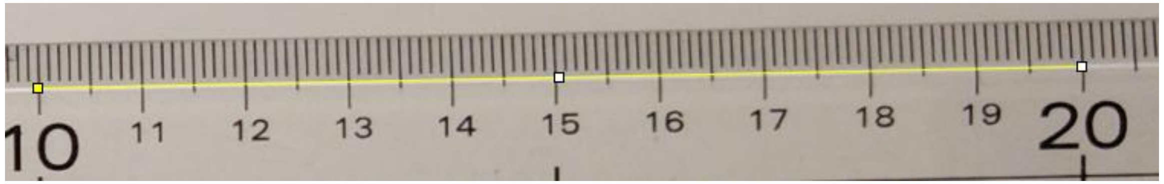
4. Draw a line along the ruler's markings

5. Analyze

6. Set scale

② Print the template sheet and measure scale

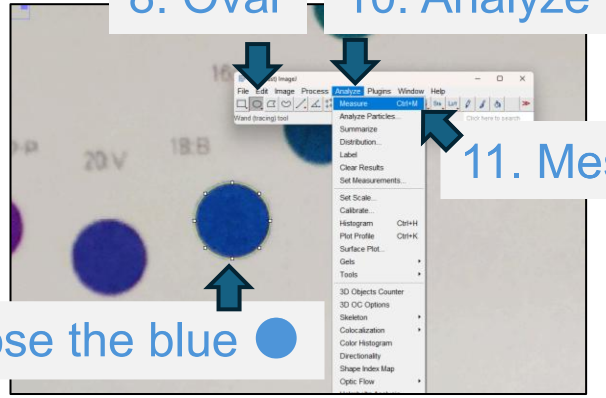




7. Set scale length with **mm**. -> OK.

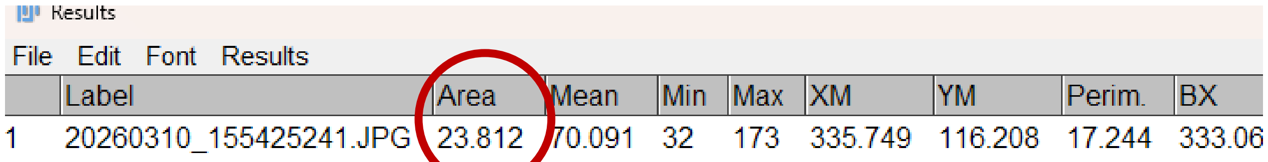
8. Oval

10. Analyze



11. Mesure

9. Enclose the blue ●



12. Make a note of this value

③ Take picture

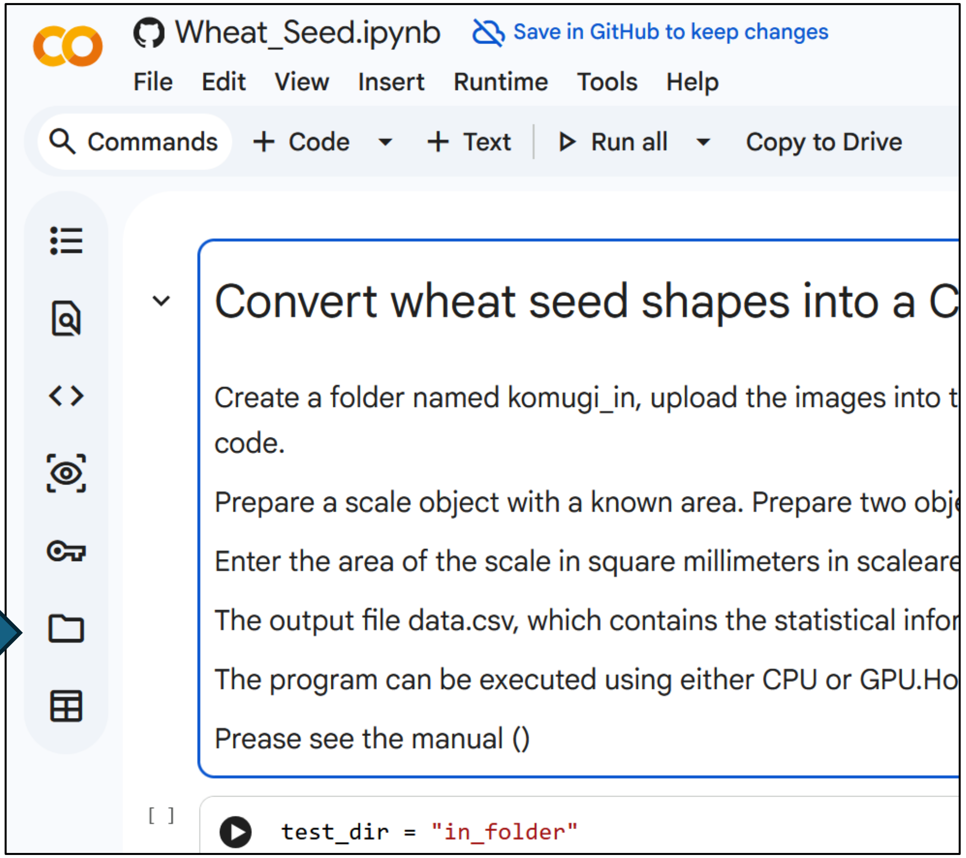
Make sure the camera is level.

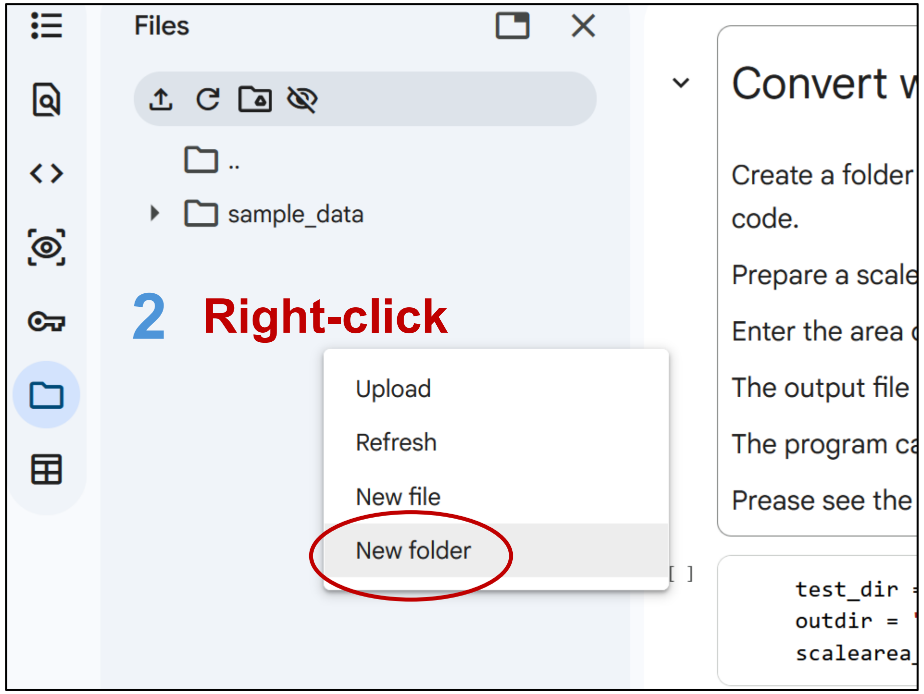
Place the seeds on a petri dish or similar dish and photograph them at the highest possible magnification.
Even if the magnification varies between samples, this is corrected by the scale.

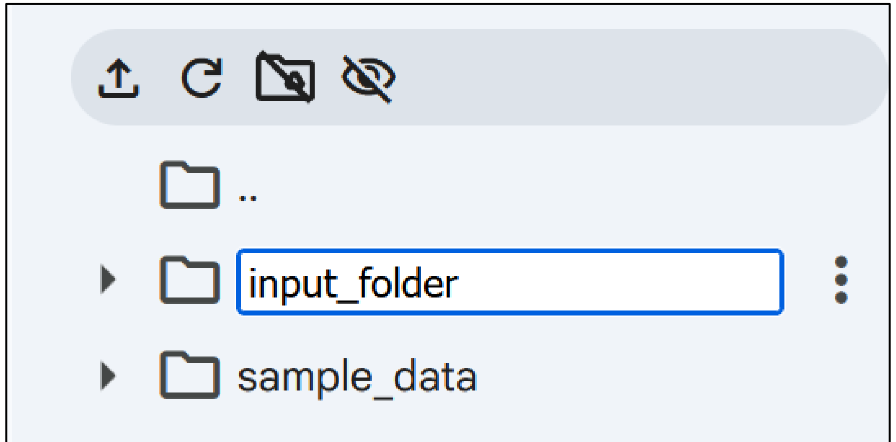
④ Run the script on google drive

Please refer to the sample data chapter and **configure it for the T4 GPU**. If the number of images is large, you may exhaust Google Colab's free quota. In that case, either purchase Compute units or download the script to your local environment and run it in Jupyter Lab. With 100 Compute units, you can analyze thousands to tens of thousands of images; upload speed will be the rate-limiting factor.

1



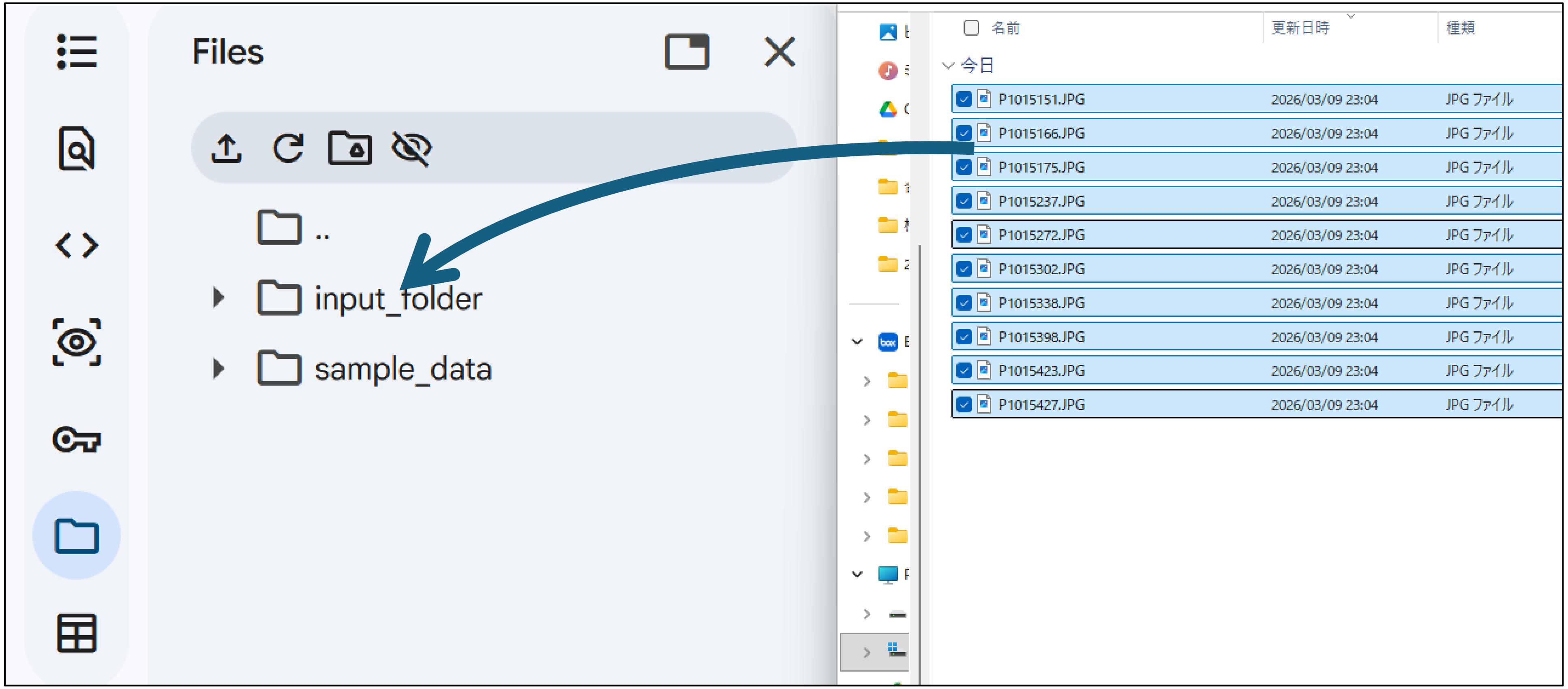




3 **(Spelling is important)**
Change folder name to “input_folder”

④ Run the script on google drive

4. Drag and drop pictures into “input_folder”



Uploading a large number of images takes considerable time,
So we recommend uploading them in batches of around 100 images.

④ Run the script on google drive

Results

File Edit Font Results

	Label	Area	Mean	Min	Max	XM	YM	Perim.	BX
1	20260310_155425241.JPG	23.812	70.091	32	173	335.749	116.208	17.244	333.06

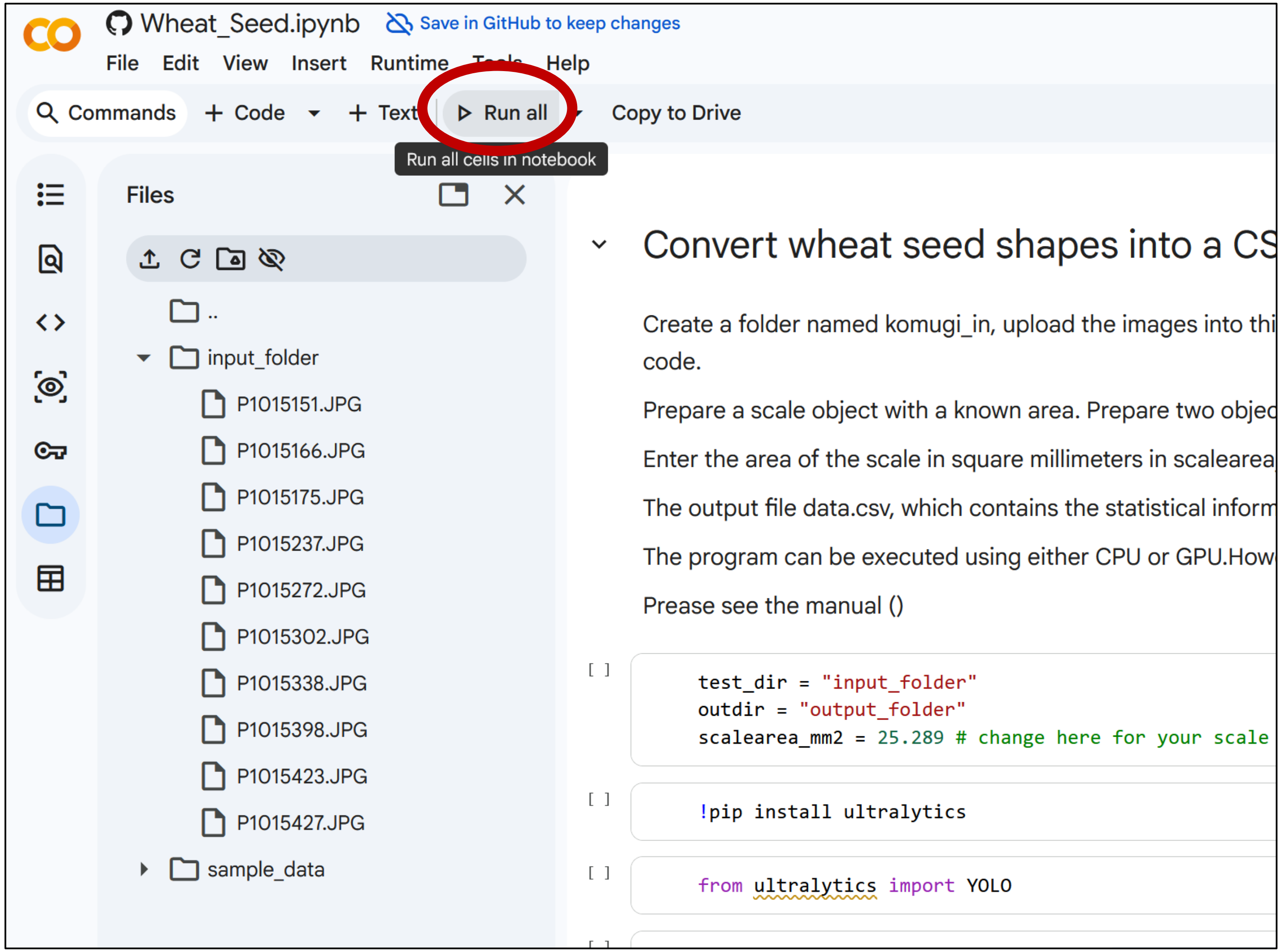
```
test_dir = "input_folder"
outdir = "output_folder"
scalearea_mm2 = 23.812 # change here for your scale
```

5. Change here to scale size

④ Run the script on google drive

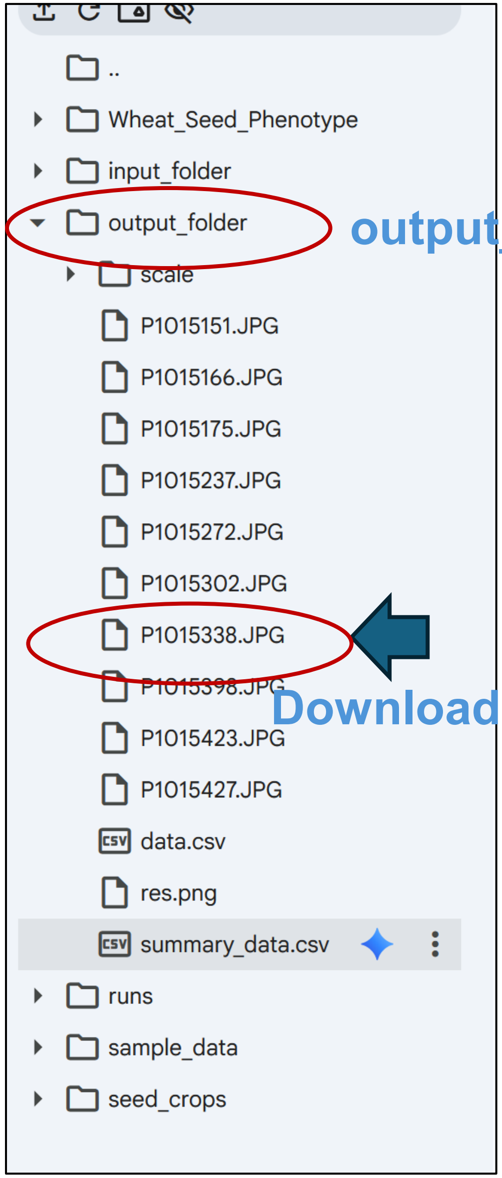
6. ▷ Run all

Please wait few minutes.



④ Run the script on google drive

7. Scroll down and wait for the cell for "Making summarized data" to complete.



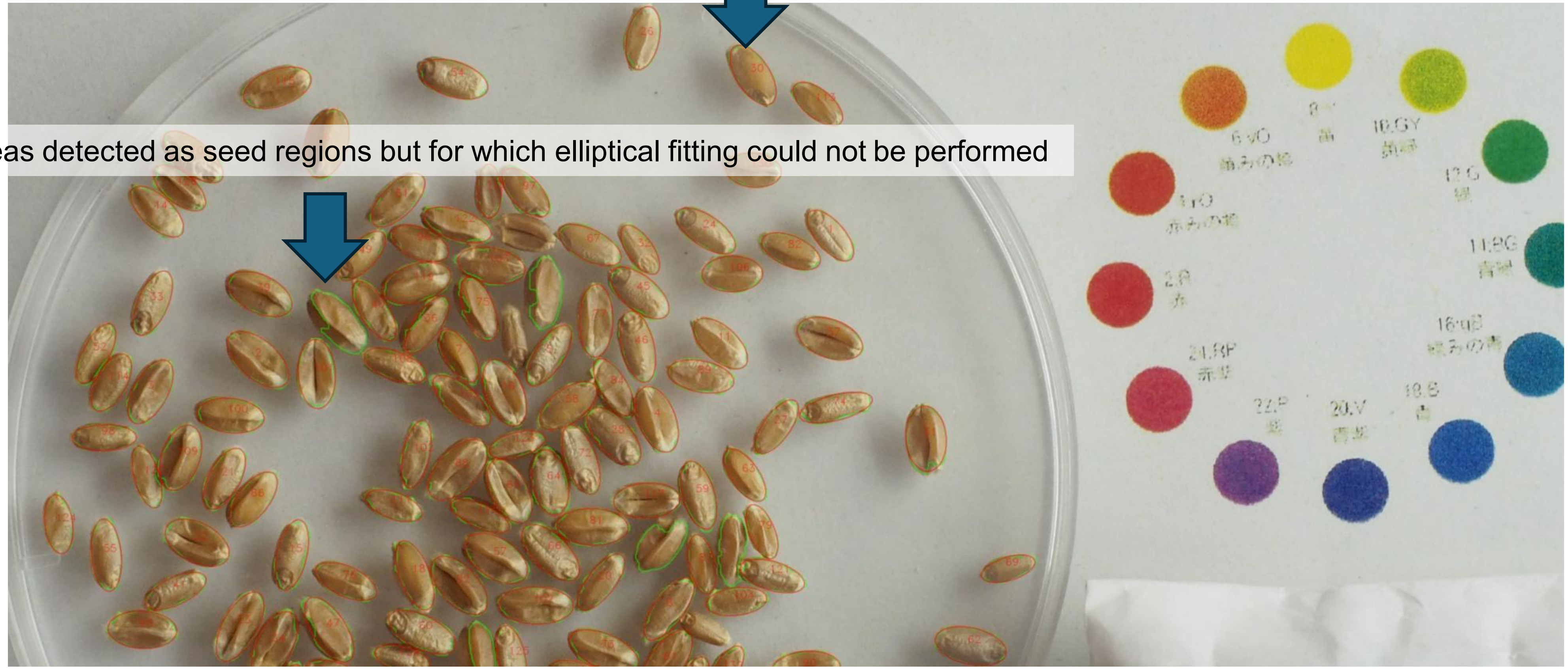
output_folder

Download picture from output_folder

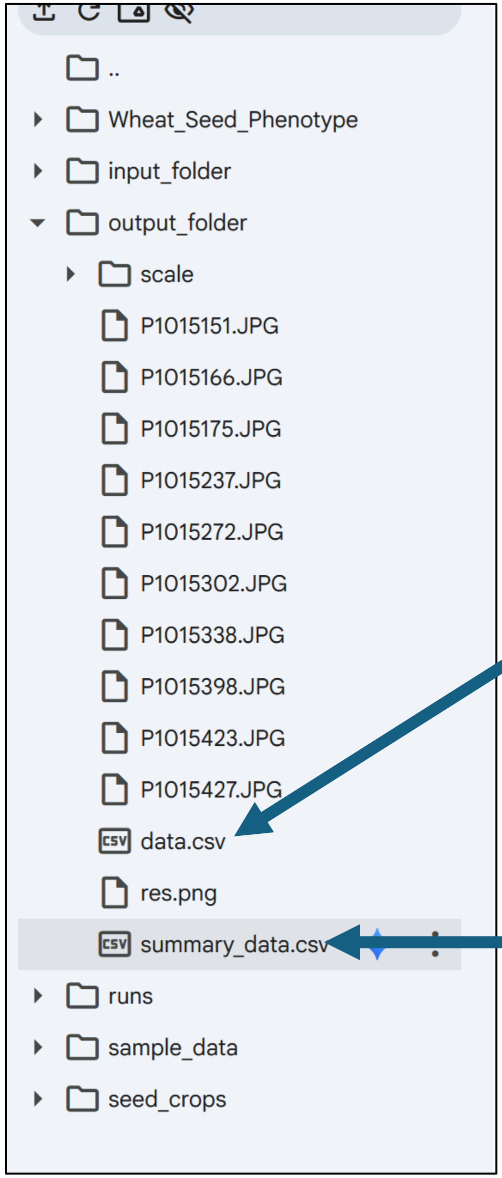
④ data check and analyze

Regions detected as seed regions where elliptical fitting was successful

Areas detected as seed regions but for which elliptical fitting could not be performed



④ data check and analyze



The Objective ID corresponds to the numerical value written within the image.

File_Name	Object_ID	Center_X	Center_Y	Major_Axi	Minor_Axi	Angle	Ellipse_Ar	scale_px	px/mm	length_mn	width_mm	Ellipticity	Area_mm2
P1015302	1	1940.738	1001.328	153.1964	77.22924	128.3857	9292.234	16945.75	25.88598	5.918122	2.983439	0.504119	13.86727
P1015302	2	1871.732	1633.051	151.3884	78.98499	155.3967	9391.326	16945.75	25.88598	5.848276	3.051266	0.521738	14.01515
P1015302	3	741.5072	1371.947	147.9797	70.9954	142.1135	8251.298	16945.75	25.88598	5.716597	2.74262	0.479764	12.31383
P1015302	4	1650.505	1524.151	152.1355	82.4005	163.5266	9845.781	16945.75	25.88598	5.877138	3.18321	0.541626	14.69336
P1015302	5	1731.345	988.1266	168.6365	93.16708	52.97364	12339.68	16945.75	25.88598	6.514588	3.599133	0.552473	18.41513
P1015302	6	798.6462	895.4789	150.9453	80.94118	52.41896	9595.75	16945.75	25.88598	5.83116	3.126835	0.536229	14.32022
P1015302	7	1058.788	1132.315	143.2501	80.79838	136.7077	9090.495	16945.75	25.88598	5.533889	3.121318	0.564037	13.56621
P1015302	8	918.8369	1016.898	154.2676	93.95677	40.91505	11383.95	16945.75	25.88598	5.959506	3.62964	0.60905	16.98884

This data contains individual seeds data

Values in millimeters are here

This data contains aggregated data for each image.